

Deborah Moyaki

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Research Focus

My research examines how faculty and students integrate technology in learning environments, focusing on faculty development, student engagement, curriculum innovation, and cross-cultural learning. By exploring digital adoption through a qualitative lens, I provide insights to enhance faculty and student support programs and leadership development opportunities. Despite the increasing investment in digital learning tools, many institutions struggle with effective adoption due to cognitive, affective, and institutional barriers. This gap results in uneven implementation, digital equity challenges, and missed opportunities for workforce preparation. To address this, I investigate technology adoption in engineering education, focusing on the lived experiences of faculty and students as they integrate emerging technologies such as Artificial Intelligence and Virtual Reality for competency-based learning. By analyzing adoption processes and behaviors, my work provides data-driven insights to optimize technology-enhanced instruction, improve digital equity, and enhance engineering workforce readiness. My research has been recognized with the ASEE Best Paper Award and is supported by NSF-funded projects; my findings inform policy development, curriculum innovation, and the strategic implementation of educational technology in engineering education.

Research Interests

Qualitative Inquiry in STEM | Technology Adoption & Acceptance Process | Emerging Technology Learning Environments | Digital Equity in Technology-Mediated Learning | Socio-technical Thinking

Research Expertise

Reflexive Thematic Analysis | Hybrid Analysis (Deductive + Inductive) | Phenomenology | Mixed Methods

Education

University of Georgia, Athens, Georgia

Ph.D. Engineering Education (CGPA 4.0/4.0)

Aug. 2022 – Dec 2026 (Expected)

Advisor: Dr. [Nathaniel Hunsu](#)

University of Ilorin, Ilorin, Kwara, Nigeria

BSc. Ed. Educational Technology

Oct. 2014 – Sept. 2018

Skills & Proficiencies

Curriculum Development, Teamwork, Assessment & Evaluation, Strategic Communication, Problem-solving, and Ability to work independently with minimal supervision.

Programming Languages and Tools: Qualtrics, SPSS, Nvivo, Microsoft Office Suite.

Publications and Conference Proceedings

Peer-Reviewed Journal Articles (Selected)

- Rukangu, A., Morelock, J., Kyle, J., & **Moyaki, D.** (2025). An Investigation into Peer-Assisted Learning in an Online Lab Environment. *Computers in Education Journal*
- **Moyaki, D.**, Hunsu, N., May, D., Paul, A., Irukulla, P., & Gomillion, C. T. (2025). Introduction of a Virtual Reality Laboratory in Tissue Engineering Course. *Biomedical Engineering Education*. (Accepted)
- **Moyaki, D.**, Oloniyo, O., Fabiyi, D., & Hunsu, N. (2025). Expectancy-Value Motivations and Student Self-Assessment in Undergraduate Engineering Learning. *Journal of Engineering Education*. (Submitted)

Book Chapters

- **Moyaki, D.**, Dunmoye, I., May, D. (2024). Students' Perceived Usefulness of Virtual Reality Laboratories. In: May, D., Auer, M.E., Kist, A. (eds) *Online Laboratories in Engineering and Technology Education*. Lecture Notes in Networks and Systems, vol 1135. Springer, Cham. https://doi.org/10.1007/978-3-031-70771-1_23

Conference Proceedings (Selected)

- **Moyaki, D.**, Hunsu, N. (2026, ASEE). Faculty Lived Experiences with Emerging Technologies for Student Competencies; a Case Study of AI. (Abstract accepted)
- **Moyaki, D.**, Dunmoye, I., May, D., & Hunsu, N. (2024, October). Student Satisfaction in Virtual Reality Laboratories. In 2024 IEEE Frontiers in Education Conference (FIE) (pp. 1-7). IEEE.
- **Moyaki, D.**, Dunmoye, I. D., Gomillion, C. T., May, D., & Hunsu, N. (2024, June). Perception of Students in Virtual Laboratories: The Role of Context. In *2024 ASEE Annual Conference & Exposition*.

- Osunbunmi, I. S., Cutler, S., Dansu, V., Brijmohan, Y., Bamidele, B. R., Udosen, A. N., **Moyaki, D.**, ... & Shih, B. P. J. (2024, June). Board 45: Generative Artificial Intelligence (GAI)-Assisted Learning: Pushing the Boundaries of Engineering Education. In *2024 ASEE Annual Conference & Exposition*.
- Jarvie-Eggart, M., May, D., **Moyaki, D.**, Carlson, K., & Smith, L. (2023, October). Co-sharing secondary qualitative research data to understand technology adoption in engineering education courses. In *2023 IEEE Frontiers in Education Conference (FIE)* (pp. 1-5).
- **Moyaki, D.**, May, D., Hunsu, N., Irukulla, P., & Gomillion, C. T. (2023, June). Introduction of a Virtual Reality Laboratory in a Tissue Engineering Course. In *2023 ASEE Annual Conference & Exposition*.

Research Experience

University of Georgia Athens, Georgia

Aug. 2023 – Present

Graduate Research Assistant

Advisor: Dr. [Nathaniel Hunsu](#)

- Led the research design, instrument selection, and approval process for a graduate research experience survey, launched in Fall 2025, to collect data from 200+ students on mental health, engagement, and research productivity for institutional policy improvements.
- Designed and conducted a comprehensive scoping review on engineering students' technology adoption behaviors, synthesizing insights from 60+ studies to identify research gaps and inform future research directions.
- Developed conceptual frameworks and implemented mixed-method research approaches to study technology-enhanced learning adoption in STEM education.
- Led manuscript development for a scoping review on technology acceptance in engineering education, refining research questions, literature reviews, and methodological alignment for high-impact publication.
- Mentored undergraduate and graduate researchers, guiding them in data collection, qualitative analysis, and literature review methodologies for technology-enhanced learning projects.
- Conducted peer reviews for ASEE, IEEE Frontiers in Education (FIE), and IJET, ensuring research integrity and quality in engineering education publications.

University of Georgia Athens, Georgia

Aug. 2022 – Jul. 2023

Graduate Research Assistant

Advisor: Dr. [Dominik May](#)

- Led research on virtual reality (VR) laboratories in biomedical engineering education, assessing their effectiveness in technical competency development and student engagement.
- Designed and implemented 5 VR laboratory modules in an undergraduate biomedical engineering course, leading to measurable improvements in student comprehension and career preparedness.
- Developed and administered structured surveys and interview protocols to investigate engineering students' experiential learning in VR labs, uncovering key trends in digital accessibility and technology adoption across diverse demographics.

Workshops & Presentations

- Reflexive Thematic Analysis – Engineering Education and Transformations Institute Research Incubator, UGA (2024)
- Deductive vs. Inductive Thinking in Categorical Modes – Engineering Student Learning and Engagement Research Lab, UGA (2024)
- A Phenomenological Approach to Technology Adoption Research in STEM – Invited Talk, Purdue University (Upcoming, Jan. 2025)

Teaching Experience

Instructor of Record

Jan 2025 – May 2025

School of Chemical, Mechanical, and Biomedical Engineering, University of Georgia

- Engineered systems in society – MCHE 2990

Instructor

Ekwuoma College Ekwuoma, Delta, Nigeria .

Nov. 2018 – Nov. 2019

- Information and Communication Technology, Basic Technology, Reading and Literacy.

Student Engagement & Leadership Experience

- Student Ambassador, Engineering Education and Transformations Institute (2025 – Present)
- Student Chapter Chair, American Society of Engineering Education, Student Division (2024 – Present)

Led a 30-member organization, organizing research webinars and networking events that increased student engagement by 50%.

- Treasurer, American Society of Engineering Education, UGA Chapter (2023 – Present)

Managed financial planning and budgeting, ensuring efficient resource allocation for student-led research activities.

- Executive board member, African Engineering Educators Forum (2023 – Present)

Facilitated international collaboration on STEM education reform in Africa, contributing to policy discussions and curriculum development.

- Professional Development Chair, Graduate and Professional Scholars (2023 – 2024)

Organized career panels and networking resources from industry and academic experts, benefitting 20+ graduate students.

Mentorship & Outreach

• National Society of Black Engineers Jr. Outreach Athens, Georgia.

Student Mentor

2023 - present

Mentoring elementary school students on STEM career paths at Barnett Shoals Elementary in Athens, Georgia.

Coordination of general body meetings centered around self-discovery and team-building activities.

Raising awareness and encouraging participation in regional and national conferences for networking and STEM career opportunities.

Student Volunteer

National Society of Black Engineers Region III Conference Atlanta, Georgia

Convention Volunteer

Nov. 2023

- Coordinated and monitored session rooms, greeting speakers, assisting exhibitors, and ensuring smooth technical operations at the National Society of Black Engineers regional conference, an annual gathering of black engineers to foster networking, academic excellence, and community impact.

Association of Visiting Secretaries Agege, Lagos, Nigeria

Financial Secretary and Academic Advisor

Jan. 2020 – Present

- Mentor post-secondary education students on career paths and college environment navigation. in preparation for examinations.
- Organize workshops on professional development for post-secondary and college students to support their educational needs.

Professional Service

Reviewer:

- ASEE American Society of Engineering Education Conference 2024 - present
- IJET International Journal of Emerging Technologies Journal 2023 to present
- FIE Frontiers in Education Conference 2023, 2024, 2025
- AERA American Educational Research Association 2023, 2025

Professional Membership

- ASEE Student Member
- AERA Student Member
- Professional Organizational Development (POD) Network
- IEEE Education Society

Student Membership

- Member, Equity Engineers Council UGA. Nov. 2023 – Present
- Member, Engineers Without Borders UGA, Sept. 2023 - Present
- Member, Graduate and Professional Scholars, March 2023 – Present
- Member, National Society of Black Engineers UGA, Jan. 2023 - Present
- Member, Society of Women Engineers UGA, Jan. 2023 to Present
- Member, African Graduate Students Forum, August 2022 to Present

Awards and Scholarships

- Student Wellbeing Initiative Grant (#2800)

Lead writer of the grant on behalf of the program student body.

- IEEE Education Society Travel Grant 2025 (\$1500)
- Rowan Future Teacher Award (\$5000)

Recognized for innovation and potential for teaching in engineering education

- Dean's Engineering Education Fellowship Award 2024

One of 5 recipients selected and funded for an academic year for excellence in engineering education research.

- POD Network Career Development Grant 2024 (\$1800)

Funded to attend and participate in the POD Network conference based on interest and contribution to educational development.

- Best Paper Award: ASEE Biomedical Engineering Division 2023

Recognized for outstanding contributions to the field of biomedical engineering education.

- Engineering Education and Transformative Institute Travel Grant Award. 2023 - 2025 (\$2500 each)

Funded to present research on technology-enhanced learning at leading global engineering education conferences